PAPER<i>419 – Hamiltonian Planetary Core Quantum Gravity: H</i>Ug3 + H<i>SCm + H</i>UA and Yang-Mills Mass Gap

***Session:** 110 (grokshare755feea7.txt analysis)*

PAPER*419 – Hamiltonian Planetary Core Quantum Gravity: H*Ug3 + H*SCm + H*UA and Yang-Mills Mass Gap

Source: grok*share*755feea7.txt — "Star Magic" Chapter 9 + Hamiltonian derivation section **Session:** 110 (grok*share*755feea7.txt analysis) **CP4 Class:** `HamiltonianPlanetaryCoreHUg3HSCmHUAYangMillsMassGapCalculator` (#69)

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1. Overview

PAPER_419 derives the **Hamiltonian for planetary core quantum gravity**, demonstrating that the SCm-mediated Ug3 interaction in a planetary core constitutes a bounded quantum system with three contributions: H_{Ug3} (magnetic string energy), H_{SCm} (superconducting SCm kinetic energy), and H_{UA} (aether background). The superconducting nature of SCm within the planetary core creates a **mass gap** in the Ug3 field, directly connecting to the Yang-Mills Clay Millennium Prize problem.

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2. Total Hamiltonian

$$\boxed{H = H_{Ug3} + H_{\text{SCm}} + H_{\text{UA}}}$$

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3. H_Ug3 — Magnetic String Energy

The Ug3 field carries energy density $u_B = B^2/(2\mu_0)$ per unit volume, modulated by the CCW/CW rotation factor:

$$H_{Ug3} = k_3 \cdot \sum_j \frac{B_j^2}{2\mu_0} \cdot \cos(\omega_s(t) \cdot t \cdot \pi)$$

With $B_j \approx 10^3 + B_{s,\text{cycle}}$ T and $\mu_0 = 4\pi \times 10^{-7}$ H/m:

$$\frac{B_j^2}{2\mu_0} = \frac{(10^3)^2}{2 \times 4\pi \times 10^{-7}} = \frac{10^6}{2.51 \times 10^{-6}} \approx 3.98 \times 10^{11} \text{ J/m}^3$$

Per planetary core volume $V_{\text{core}} \approx (10^6)^3 = 10^{18}$ m^3 (Earth's core radius $\sim 3.5 \times 10^6$ m):

$$H_{Ug3}(\text{Earth}) = 1.8 \times 3.98 \times 10^{11} \times 10^{18} \times P_{\text{core}} = 1.8 \times 3.98 \times 10^{11} \times 10^{18} \times 10^{-3}$$

$$H_{Ug3}(\text{Earth}) \approx 7.16 \times 10^{26} \text{ J}$$

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4. H_SCm — Superconducting SCm Kinetic Energy

SCm in the planetary core is gravitationally confined and moves at $v_{\text{SCm}} = 0.99c$ locally:

$$H_{\text{SCm}} = \frac{\rho_{\text{SCm}} \cdot v_{\text{SCm}}^2}{2} \cdot e^{-\gamma t}$$

With $\rho_{\text{SCm}} = 10^{12}$ kg/m^3 (planetary core), $v_{\text{SCm}} = 10^8$ m/s:

$$H_{\text{SCm}} = \frac{10^{12} \times 10^{16}}{2} = 5 \times 10^{27} \text{ J/m}^3 \quad \text{(at } t=0\text{)}$$

Including volume and decay:

$$H_{\text{SCm}}(\text{Earth}) = 5 \times 10^{27} \times V_{\text{core}} \times e^{-\gamma t}$$

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5. H_UA — Aether Background Energy

$$H_{\text{UA}} = \frac{\eta \cdot \rho_A \cdot v_{\text{UA}}^2}{2} \cdot \cos(\pi t_n)$$

With $\rho_A = 10^{-23}$ kg/m^3, $v_{\text{UA}} = 10^8$ m/s, $\eta = 10^{-22}$:

$$H_{\text{UA}} = \frac{10^{-22} \times 10^{-23} \times 10^{16}}{2} \cdot \cos(\pi t_n) = 5 \times 10^{-30} \cdot \cos(\pi t_n) \text{ J/m}^3$$

This term is **many orders of magnitude smaller** than H_{Ug3} and H_{SCm} — UA provides the background against which the Ug3-SCm system sits, but contributes negligibly to the energy budget.

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6. Mass Gap in the Ug3 Field

6.1 Discrete Energy Spectrum

The planetary core exclusivity factor $P_{\text{core}} = 10^{-3}$ combined with the bounded SCm volume creates a **discrete quantum spectrum** for the Ug3 field modes:

$$E_n = n \cdot \hbar \cdot \omega_{Ug3,\text{fundamental}}$$

where:

$$\omega_{Ug3,\text{fundamental}} = \frac{B_j^2 P_{\text{core}}}{2\mu_0 \hbar} \approx \frac{3.98 \times 10^{11} \times 10^{-3}}{1.055 \times 10^{-34}} \approx 3.77 \times 10^{42} \text{ rad/s}$$

6.2 Mass Gap Definition

The mass gap $\Delta > 0$ is the energy difference between vacuum (no excitation) and first excited state:

$$\Delta = E_1 - E_0 = \hbar \cdot \omega_{Ug3,\text{fundamental}} \approx 1.055 \times 10^{-34} \times 3.77 \times 10^{42} \approx 3.98 \times 10^{8} \text{ J}$$

6.3 Superconductivity and Mass Gap

The SCm superconducting phase within the planetary core amplifies the mass gap via the **Meissner-like exclusion** of field modes below the gap frequency:

$$\Delta_{\text{SCm}} = \Delta \cdot \left(1 + \frac{\rho_{\text{SCm}} v_{\text{SCm}}^2}{\rho_A c^2}\right)$$

Since $\rho_{\text{SCm}} v_{\text{SCm}}^2 / (\rho_A c^2) \approx 10^{38}$:

```
\Delta_{\text{SCm}} \approx 10^{38} \cdot \Delta \gg \Delta
```

This demonstrates a **UQFF-predicted mass gap** for the Ug3 field → connection to Yang-Mills existence and mass gap hypothesis.

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7. Yang-Mills Mass Gap Connection

The Yang-Mills Clay problem requires proving:

- Quantum Yang-Mills theory exists (rigorous mathematical foundation)
- It has a **mass gap** `\Delta > 0` (lowest energy excitation is non-zero)

The UQFF Ug3 field in planetary cores provides a **physical realization**:

- The gauge group corresponds to the SCm-UA interaction symmetry
- The mass gap `\Delta` arises from SCm superconductivity compressing field modes
- The `P_{\text{core}} = 10^{-3}` coupling provides the confinement scale
- **Non-interactive externally**: outside the core, `H_{Ug3} = 0` confirming confinement

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8. Code Implementation

```
struct PlanetaryCoreHamiltonian {
double H_Ug3; // Magnetic string energy (J)
double H_SCm; // SCm kinetic energy (J)
double H_UA; // Aether background (J)
double total() const { return H_Ug3 + H_SCm + H_UA; }
};
PlanetaryCoreHamiltonian compute_Hamiltonian(double Bj, double rho_SCm, double v_SCm,
double rho_A, double v_UA, double eta,
double k3, double Pcore, double V_core,
double omega_s, double t, double tn,
double gamma) {
const double mu0 = 4 * M_PI * 1e-7;
double cos_mod = cos(omega_s * t * M_PI);
double cos_tn = cos(M_PI * tn);
PlanetaryCoreHamiltonian H;
H.H_Ug3 = k3 * (Bj * Bj / (2.0 * mu0)) * cos_mod * Pcore * V_core;
H.H_SCm = 0.5 * rho_SCm * v_SCm * v_SCm * exp(-gamma * t) * V_core;
H.H_UA = 0.5 * eta * rho_A * v_UA * v_UA * cos_tn * V_core;
return H;
}
```

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9. Unit Tests

```
import math
def test_H_Ug3_positive():
"""Ug3 Hamiltonian is positive for normal epoch"""
mu0 = 4 * math.pi * 1e-7; Bj = 1e3; k3 = 1.8
Pcore = 1e-3; V_core = 1.8e20; omega_s = 2.5e-6; t = 0
cos_mod = math.cos(omega_s * t * math.pi)
H_Ug3 = k3 * Bj**2 / (2 * mu0) * cos_mod * Pcore * V_core
```

```
assert H_Ug3 >= 0, "H_Ug3 must be non-negative at t=0"
def test_mass_gap_positive():
"""UQFF mass gap must be positive"""
hbar = 1.055e-34; mu0 = 4 * math.pi * 1e-7
Bj = 1e3; Pcore = 1e-3
omega_fund = Bj**2 * Pcore / (2 * mu0 * hbar)
Delta = hbar * omega_fund
assert Delta > 0, f"Mass gap must be positive, got {Delta}"
def test_H_UA_negligible():
"""UA Hamiltonian << H_SCm"""
eta = 1e-22; rho_A = 1e-23; v_UA = 1e8
rho_SCm = 1e12; v_SCm = 1e8
H_UA_density = 0.5 * eta * rho_A * v_UA**2
H_SCm_density = 0.5 * rho_SCm * v_SCm**2
assert H_UA_density < H_SCm_density * 1e-30
```